

Lesson 4 · A Stateful Agent with LangGraph

PDF: [this lesson](#) · **Install (Linux · macOS · Windows):** [guide](#) · [PDF](#)

Part of **local-ai-lab** — a hands-on course for building local AI.

Course home: <https://nikolareljin.github.io/local-ai-lab/> **Source:** <https://github.com/nikolareljin/local-ai-lab>

Lessons: 1 · [RAG](#) → 2 · [MCP](#) → 3 · [LangChain](#) → 4 · **LangGraph (you are here)** → 5 · [Ollama tools](#) → 6 · [Semantic Kernel](#) → 7 · [Bedrock Agents](#) → 8 · [Google ADK](#)

Status: planned. Outline below; full step-by-step coming later. □ the repo to follow along.

The idea

A straight RAG chain (Lessons 1 and 3) is **linear**: retrieve → ground → answer, once. Real assistants need **control flow** — decide whether to retrieve at all, retry with a reformulated query when results are weak, call more than one tool, and remember the conversation.

LangGraph models this as a **stateful graph** of nodes and edges with explicit state.

This lesson turns the RAG pipeline into an agent that **thinks about its own retrieval**.

What you'll build

A graph with nodes such as:



What you'll learn

- **State** — a typed graph state that flows between nodes (question, docs, draft, attempts)
- **Nodes & edges** — retrieval, grading, query rewriting, and generation as composable steps
- **Conditional edges** — branch on "are these results good enough?" and loop to self-correct
- **Cycles** — corrective RAG: re-retrieve with a better query instead of answering badly

- **Memory & checkpoints** — persist state so a conversation survives across turns
- **Tool routing** — combine the Lesson 2 MCP `search_docs` tool with other tools

Builds on the whole course

Concept	Comes from
retrieve / ground / answer	Lesson 1
<code>search_docs</code> as a callable tool	Lesson 2
LangChain retrievers & chat models	Lesson 3
graph, state, cycles, self-correction	Lesson 4 (this one)

Where it all lands: by the end of the course you can build a local, private, self-correcting document agent — and explain every layer, from `pypdf` extraction up to a LangGraph control loop.

Prerequisites

Lessons [1-3](#). The agent reuses the retriever, the MCP tool, and the LangChain components from earlier lessons.

Next lesson

Lesson 5 · Ollama + Function Calling → — give a local model its own tools, fully offline, then carry the same `search_docs` capability through Semantic Kernel, Bedrock, and ADK.

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